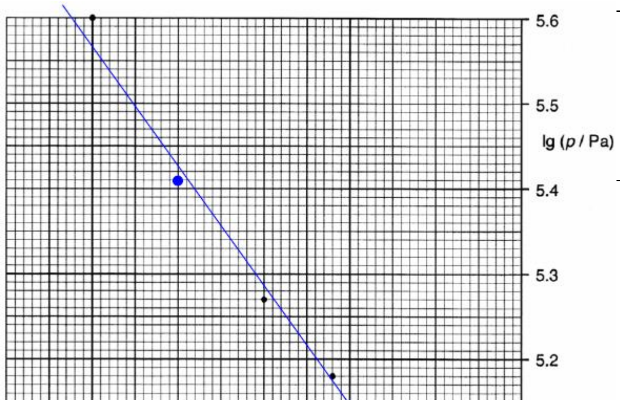
8(a)	Show with at least 2 sets of values that p is not equals to k /V, ie. $p_1V_1 \neq p_2V_2$ 1 mark deduced for any incorrect reading of the 2 sets of data	B2
(b)(i)	pV = nRT. From the graph, pV increases when the gas is compressed. Therefore temperature increases.	M1 A1
(ii)	Moving piston transfers K.E to the argon atoms thereby <u>rebound speed</u> becomes <u>higher</u>, and this <u>increases the mean K.E of argon atoms, thus temperature increases.</u>	B1
(c) (i)	From Fig. 7.2, when volume $V = 1.00 \times 10^{-3}$ m^3, pressure p = 2.6×10^5 /2.65×10^5 lg (*p* / Pa) = 5.41/ 5.42	A1
(ii)	Correct plot Line of best fit drawn	A1 A1



	-X = gradient Y-intercept = lg y 2 points read correctly from the graph - x = (5.0 – 5.6) / (-2.69 – (-3.12)) x = 1.4 Correct substitution and determination of x Substitute x = 1.4 into the straight line equation, using the point (-2.69, 5.0) 5.0 = -1.4(-2.69) + lg y lg y = 1.234 y = 17 Correct substitution and determination of y	B1 A1 A1 A1
d (i)	$19.5\% \times 3.0 \times 10^{23} = 5.85 \times 10^{22}$	A1
(ii)	(Percentage < 200 m s^{-1}) = $1\% + 2.5\% = 3.5\%$ $3.5\% \times 3.0 \times 10^{23} = 1.05 \times 10^{22}$	A1
(iii)	Total percentage = 98.5% $1.5\% \times 3.0 \times 10^{23}$ $= 4.5 \times 10^{21}$	C1 A1
(iv)	$C_{rms} = \sqrt{\dfrac{n_1 v_1^2 + n_2 v_2^2 + \ldots . n_N v_N^2}{n_1 + n_2 + \ldots .. n_N}}$ $= \sqrt{\dfrac{1 \times 50^2 + 2.5 \times 150^2 + 9.5 \times 250^2 + 18 \times 350^2 + 21.5 \times 450^2 + 19.5 \times 550^2 + 13.5 \times 650^2 + 7.5 \times 750^2 + 4 \times 850^2 + 1.5 \times 950^2}{98.5}}$ = 526 m s^{-1} Hence, C_{rms} falls in the range of 500 -600 m s^{-1} Note : Concept needs to be correct to obtain the last mark for the range.	C1 A1 B1
(e)	Sketch showing more atoms at higher speed and fewer atoms at lower speed.	B1